5C: Consumer Search

Industrial Organization

Renmin University of China

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1 Introduction

2 Weitzman Search

3 Information Provision in Search Markets

4 Conclusion

Consumer Search

- Most of our lectures abstract away from sophisticated consumer actions.
  - Consumers are typically passive players or employ trivial strategies.
- Some common and important consumer actions in reality:
  - Learning (as discussed in 3B)
  - Search

Outlines

This lecture note covers two papers on consumer search.

- *Optimal Search for the Best Alternative* (Weitzman, **Econometrica**, 1979)
  - Weitzman search
  - Covered by Section 5.1 in *Platfrom Economics* (written by ours)
- *Attraction versus Persuasion: Information Provision in Search Markets* (Au and Whitmeyer, **JPE**, 2023)
  - Weitzman search + information design

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Setup: Pandora's Problem (IO Setting)

- Sellers $N = \{1, 2, \ldots, n\}$ indexed by i
  - Seller i has a single product with value v_i drawn from a known CDF $F_i(v_i)$.
    - Values are independent.
  - Search friction: Visiting seller i costs $c_i \geq 0$ and takes time $t_i \geq 0$.
    - The consumer learns v_i after waiting for time t_i.
    - Discount factor $\beta_i = e^{-rt_i}$
- A consumer with unit demand (expected present value maximizer)
  - Fallback value $v_0 = 0$
  - At each stage she must decide whether or not to visit a seller.
    - If stops searching, she receives the maximum value she has learned. (Game ends.)
    - Otherwise, she selects the next seller, pays the cost, and waits for the outcome. (Then comes the next stage.)
  - Present value maximizer: the search costs are paid *during* search, but the *maximum* value is collected *after* search has stopped.

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Dynamic Programming Format

- The set of visited sellers $S \subset N$ and the set of unvisited sellers $S^c = N \setminus S$
- Maximum learned value $y = \max_{i \in S \cup \{0\}} v_i$
- State of the system (S^c, y)
  - Define $\Psi(S^c, y)$ as the expected present value of following an optimal policy from this time on when the state is (S^c, y).
  - The state valuation functions Ψ satisfies

$$\Psi(S^c, y) = \max\left\{y, \max_{i \in S^c}\left\{-c_i + \beta_i\left[\Psi(S^c \setminus \{i\}, y)\int_{-\infty}^{y} \mathrm{d}F_i(v_i) + \int_{y}^{\infty} \Psi(S^c \setminus \{i\}, x_i)\mathrm{d}F_i(v_i)\right]\right\}\right\},$$

(Optimality for dynamic programming)

where $\Psi(\varnothing, y) = y$.

The Solution Method

- A dynamic programming problem is solved by backward induction.
- Last period question: Given a maximum learned value y and the distribution F_i of the remaining seller, search or stop?
  - If does not search, the value: y
  - If search, the (PV of) expected value: $\beta_i \mathbb{E}_{v_i \sim F_i}[\max(v_i, y)] - c_i$
- **Definition**: The reservation price/value z_i is the value amount that makes the consumer indifferent between
  1. Stopping search immediately and accepting z_i.
  2. Visiting seller i, and then stopping (accepting the larger of z_i or v_i).
- Mathematically, z_i is given by

$$\underbrace{c_i}_{\text{The cost of searching seller } i} = \underbrace{\beta_i \int_{z_i}^{\infty} (v_i - z_i) \mathrm{d}F_i(v_i) - (1 - \beta_i) z_i}_{\text{The expected benefit of searching seller } i \text{ given a fallback of } z_i} . \quad (1)$$

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Optimal Policy

- The optimal last-period policy: if $y \geq z_i$, stop; otherwise, search.
- Two direct implications:
  - If $y < \max_{i \in S^c} z_i$, the search should continue. (Search the seller $\arg\max_{i \in S^c} z_i$)
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Theorem 1 (Pandora's Rule)

The optimal search policy (consumer strategy) is characterized by

(i) ***Selection Rule****: The consumer searches in decreasing order of reservation prices.*

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Remarks

- We can evaluate each seller and simply rank them by reservation prices.
- Properties of reservation prices
  1. The reservation price is insensitive to the probability distribution below z_i.
  2. A riskier distribution (specifically, a *mean-preserving spread*) increases z_i.
- In Weitzman search, consumers prefer *dispersion*.
  - A seller with a low probability of a massive value might be searched before a "safe" seller, because the "safe" seller can be kept as a backup.

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Exercise

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Let v_1 and v_2 be the values of products 1 and 2. Assume that

$$\Pr(v_1 = 0) = \Pr(v_1 = 1) = \frac{1}{2}, \qquad v_2 \sim U[0, 1].$$

Search costs are $c_1 = c_2 = 1/4$ and there is no discounting. Let F_1 and F_2 be the cumulative distribution functions of v_1 and v_2. The reservation prices for products 1 and 2 are z_1 and z_2.

(a) Write down $F_1(x)$ and $F_2(x)$ for all $x \in [0, 1]$.

(b) Show that F_1 is a mean-preserving spread of F_2.

(c) Find z_1 and z_2, and show that $z_1 > z_2$.

Proof of Theorem

Proof First, the reservation prices are well defined. Let

$$H_i(z) = \beta_i \int_z^{\infty} (v_i - z) \mathrm{d}F_i(v_i) - (1 - \beta_i) z.$$

One can easily verify that $H_i(z)$ is continuous and monotonic, $H_i(-\infty) = +\infty$, $H_i(+\infty) = -\infty$ (0 if $\beta = 1$). When $c_i > 0$ or $\beta < 1$, $c_i = H_i(z_i)$ gives a unique solution z_i.

The proof is then done by **induction** on the number of unvisited sellers $m = |S^c|$. For $m = 1$, the optimality is directly implied by the definition of reservation price.

Proof of Theorem

Proof (continued) Suppose the optimality of Pandora's Rule holds for m unvisited sellers and any value of y. Consider the situation with $m+1$ unvisited sellers and any value of y. Let $j \in \text{argmax}_{i \in S^c} z_i$.

If $y \geq z_j$, after one seller is visited, by the $|S^c| = m$ case it will be optimal to stop. Thus, by comparing visiting exactly one seller and not visiting any, we know the latter option is better. The stopping rule is thus proved for $m+1$.

If $y < z_j$, just visiting j and then stopping would yield a higher expected present discounted value. Thus, at least one seller should be visited.

Proof of Theorem

Proof (continued) Suppose, by **contradiction** with Pandora's Rule, it is optimal to open $k \in S^c$ first, where $z_k < z_j$. WLOG let $z_k \in (y, z_j)$ (otherwise it follows from the $y \geq z_i$ case visiting k is worse than stopping). By the $|S^c| = m$ case, we should follow Pandora's Rule thereafter. Let the expected discounted present value of visiting k and then following Pandora's Rule be B.

Consider the following alternative. Visit j first. Let $h \in \operatorname{argmax}_{i \in S^c \setminus \{j\}} z_i$ be a seller with the second biggest reservation price. If $x_j \geq z_h$, terminate. Otherwise, visit k next and then proceed by Pandora's Rule. Let the expected present discounted value of this alternative policy be A.

The proof is completed by verifying that $A > B$.[1] $\square$

[1] This verification is purely algebraic. For details, see the original paper.

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Motivation

Firms control how much information consumers' visits will bring.

- Objective 1: Persuasion
  - Each firm wants consumers to select its product over those of its competitors.
  - Examples:
    - Car sellers choose whether to offer test drives.
    - Software companies decide the length of free trial periods.
- Objective 2: Attraction
  - Each firm wants consumers to try its product early in their search.
  - Information about these inspections are typically made available to prospective consumers before they encounter the products.
  - Examples:
    - Trial durations
    - Functional restrictions of the trial version

Persuasion-Attraction Trade-Off

- Conditional on the consumer arriving, a firm wishes to maximize the conversion rates from visits to purchases.
  - Persuasion incentive $\Rightarrow$ **concentration** of beliefs over the stopping threshold
  - An insight from information design
- Committing to provide more information improves a firm's position in the consumer's search order.
  - Attraction incentive $\Rightarrow$ **spreading** of beliefs to entice the consumer to visit
  - An insight from Weitzman search
- This paper abstracts away from price competition.

Market Setup

- Firms $N = \{1, 2, \ldots, n\}$ indexed by i, ex ante identical
  - Each seller has a single product with an uncertainty quality (or match value) to the consumer of either 0 or 1.
    - The qualities are identically and independently distributed.
    - The prior probability of the quality being 1 is $\mu \in (0, 1)$.
  - A firm receives a payoff normalized to 1 if its product is picked and 0 if not.
- A consumer with unit demand (risk-neutral)
  - No outside option

Pre-search Stage

- At the beginning, neither the consumer nor the firms know the quality realizations of the firms' products.
- Each firm simultaneously commits to a signal, $\pi_i : \{0, 1\} \to \Delta S$, with some some space of realizations S.
  - The signals are public and shape the consumer's behavior directly.
  - Mixed strategies (i.e., randomization over signals) are allowed.
- The firms' randomizations are resolved simultaneously before the consumer starts her sequential search.

Search Stage

- Guided by the informativeness of the signals, the consumer learns about the firms' product qualities by visiting the firms and observing their signal realizations in sequence.
  - Each inspection requires a search cost of $c > 0$.
- After observing the signal realization of a firm, the consumer updates her prior to the posterior expected quality of the product offered by that firm.
- Weitzman search setting
- Assume $c \leq \mu$, so that search is not strictly dominated for the consumer.

Solution Concept

- Subgame-perfect Nash equilibrium
- Given the ex ante symmetry of the firms, it is natural to focus on symmetric equilibria in which
  1. All firms adopt a common (possibly mixed) strategy.
  2. The consumer uses a tie-breaking rule that treats all firms identically.

Basics

- Bayesian persuasion $\Rightarrow$ The strategy space of each firm can be redefined to be the set of feasible distributions over posterior qualities.
  - A distribution over posteriors, F_i, is feasible if it has mean μ and its support is a subset of $[0, 1]$.
  - $\mathcal{F}$ denotes the set of feasible distributions.
- Weitzman search $\Rightarrow$ The consumer's search behavior is characterized by Pandora's Rule.
  - F_i, chosen by firm i, define its reservation value/price $U(F_i)$.
  - For any $F_i, G_i \in \mathcal{F}$, if F_i is a mean-preserving spread of G_i, then $U(F_i) \geq U(G_i)$.
  - The set of feasible reservation values

$$\{U(F_i) : F_i \in \mathcal{F}\} = [\underline{U}, \overline{U}], \quad \underline{U} = \mu - c, \quad \overline{U} = 1 - c/\mu.$$

Pure Strategy Equilibra

Lemma 1

There exist no symmetric pure strategy equilibria in which any reservation value $U < \overline{U}$ is induced.

Theorem 2 (Full Information Equilibrium)

Define $\overline{\mu} = 1 - (1/n)^{-1/(n-1)}$. A symmetric equilibrium in pure strategies exists if and only if $\mu \geq \overline{\mu}$. In this equilibrium, all firms provide full information.

Reformulation: Discrete Choice

The consumer's sequential search can be formulated as a static discrete choice problem, which makes symmetric equilibria in mixed strategies tractable (Armstrong, 2017; Choi et al., 2018).

- The effective value of firm i is defined as $W_i \equiv \min\{p_i, U(F_i)\}$, where $p_i \in [0, 1]$ is the realized posterior quality.
  - Choi et al. (2018)'s observation: Under Pandora's Rule, there are only two scenarios in which firm i makes a sale:
    1. If firm i's effective value is its reservation price, the consumer purchases from it whenever she visits it, an event that occurs if and only if all of the other firms she visited earlier realize posterior qualities below $U(F_i)$.
    2. If firm i's effective value is its posterior quality p_i, it makes a sale if and only if firms whose posterior qualities are greater than p_i are not yet visited.

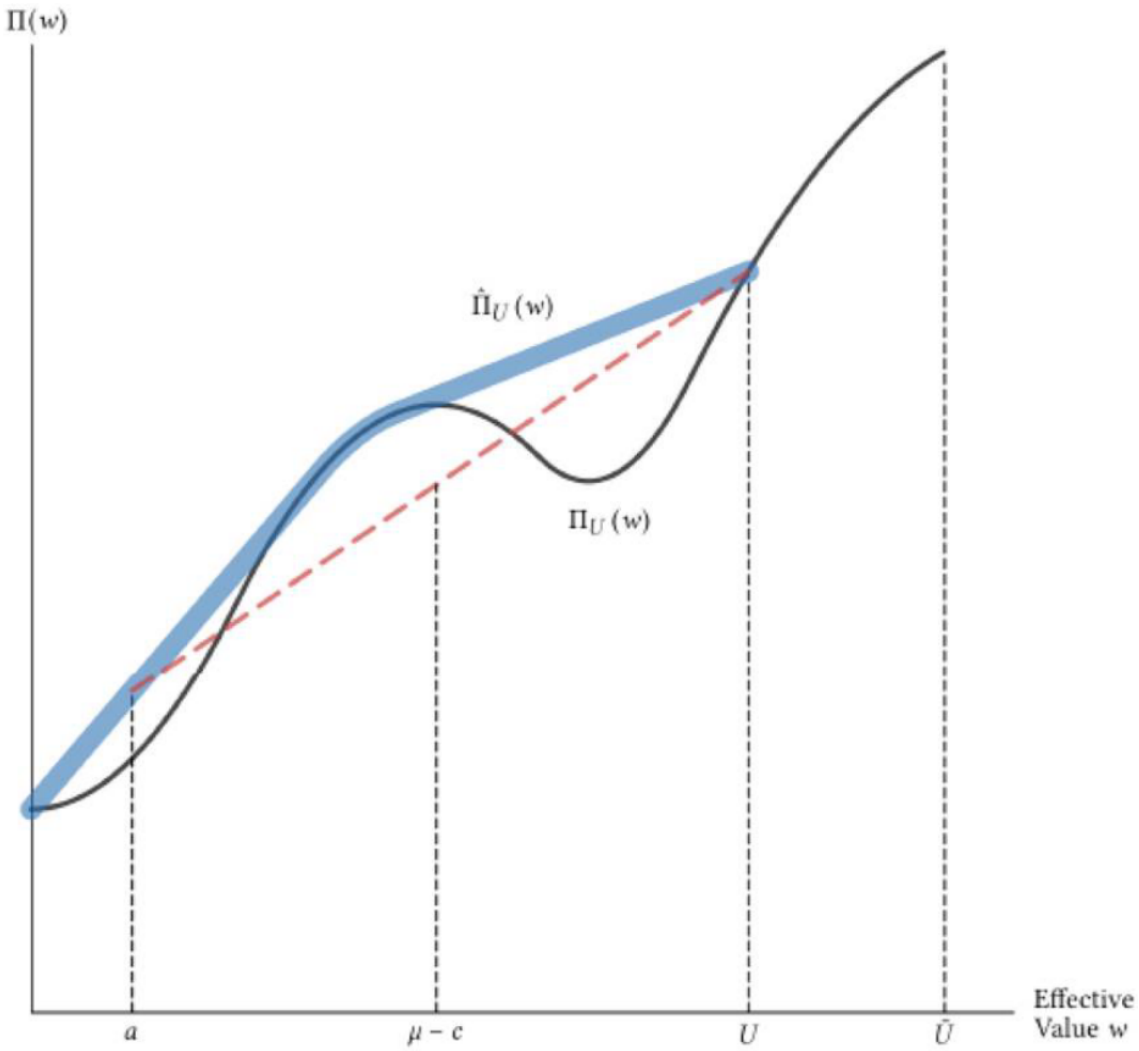
Reformulation: Discrete Choice

- The winning condition for firm i: its effective value realization beats (i) the posterior qualities of those competitors visited before it, and (ii) the reservation values of those with higher posterior qualities.
  - Firm i wins only if it has the highest effective value realization; i.e., $W_i = \max_{j=1,2,\ldots,n} W_j$.
- Two (mixed) strategies of a firm are *payoff equivalent*, as long as their implied effective value distributions are identical.
- This transformation requires us to characterize the set of effective value distributions that can be induced by a firm's strategy.

Inducible Effective Value Distribution

- An effective value distribution is inducible if and only if there is a pure or mixed strategy that can induce it.
- Harder to deal with than a standard Bayesian persuasion problem! A simple mean condition no longer exists.

Concavification



Symmetric Equilibrium

Theorem 3

Except for the case in which $\mu = 1/2$ and $n = 2$, there exists a unique symmetric equilibrium in the firms' competition over effective values. If $\mu = 1/2$ and $n = 2$, there exists a continuum of symmetric equilibria. The equilibrium effective value distribution gives rise to a payoff function with a linear structure.

Corollary 1

For $c > 0$, an increase in c worsens the equilibrium distributions over effective values in the sense of first-order stochastic dominance, thus hurting the ex ante consumer welfare.

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Conclusion

- Weitzman (1979)
  - The basic Weitzman search model
  - The optimal search policy – Pandora's Rule
  - The consumer prefers dispersion.
- Au and Whitmeyer (2023)
  - This paper extends Weitzman (1979) by endogenizing value distributions.
  - The model abstracts away from price competition.
  - The characterization combines Weitzman search and BP.